Camera-Driven Real-Time Digital Twin

Autonomous Pioneer P3DX Navigation with Physical Obstacle Synchronisation

Environment: Ubuntu 22.04 LTS · Python 3.10 · OpenCV 4.8 · CoppeliaSim Edu 4.10 (ZeroMQ API)

System Overview & Architecture

This digital twin links a physical desk observation area to a 3D virtual simulation in **CoppeliaSim**. A real camera observes physical markers or colored objects, estimates their coordinates in real time, synchronises their state into an open 3.0m × 3.0m virtual arena, and drives a simulated **Pioneer P3DX** mobile robot that navigates autonomously while reacting safely to obstacle introductions.

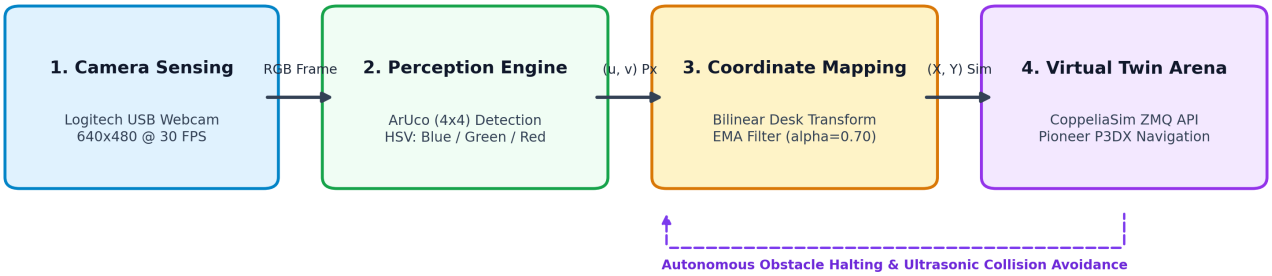


Figure 1: Real-time pipeline: physical webcam sensing → perception → spatial mapping → virtual twin.

The pipeline executes asynchronously at **30 FPS** across four dedicated stages:

1. **Physical Sensing:** A Logitech USB webcam captures RGB frames (640 × 480 @ 30 FPS) via Linux V4L2.
2. **Perception Engine:** Extracts object centroid (u_t, v_t) in pixel space with sub-millisecond execution.
3. **Spatial Transformation:** Maps pixel coordinates to simulation coordinates $(X_{\text{sim}}, Y_{\text{sim}})$ with natural desk sensitivity.
4. **Virtual Twin Synchronisation:** Updates the virtual obstacle in CoppeliaSim via ZeroMQ Remote API (< 15 ms latency).

Real-Time Perception Engine

To guarantee deterministic, high-speed tracking without neural network inference overhead, two perception modes are supported:

- **ArUco Fiducial Marker Tracking:** Detects predefined `DICT_4X4_50` markers. Four high-contrast corners are resolved to sub-pixel accuracy, computing the geometric centroid with high immunity to ambient illumination changes.
- **Three-Color Presets (HSV):** Provides three clean, mutually exclusive color filters tailored to common objects:
  - **Blue Preset [Key 1]:** $H \in [100, 135]$, $S \in [100, 255]$, $V \in [60, 255]$
  - **Green Preset [Key 2]:** $H \in [35, 85]$, $S \in [70, 255]$, $V \in [50, 255]$
  - **Red Preset [Key 3]:** Wraps hue boundary ($H \in [0, 10] \cup [168, 180]$, $S \in [100, 255]$, $V \in [70, 255]$)

  Binary masks are filtered with morphological opening and closing (5×5 ellipse) to eliminate salt-and-pepper noise before computing spatial moments ($c_u = M_{10}/M_{00}, c_v = M_{01}/M_{00}$).

Bilinear Coordinate Mapping & Sensitivity Scaling

The physical camera monitors an active desk tracking zone:

$$u \in [u_{\min}, u_{\max}] = [100, 540] \text{ px}, \quad v \in [v_{\min}, v_{\max}] = [60, 420] \text{ px} \tag{1}$$

Raw coordinates are normalized $\bar{u}, \bar{v} \in [-1.0, 1.0]$. A sensitivity gain ($G = 1.0$) maps natural desk hand movements directly across the 3.0m × 3.0m virtual arena ($X, Y \in [-1.4\text{m}, +1.4\text{m}]$), preventing the object from inadvertently exiting the camera frame. An Exponential Moving Average (EMA) filter eliminates high-frequency hand jitter:

$$\mathbf{p}_{\text{filtered}}[t] = \alpha \cdot \mathbf{p}_{\text{raw}}[t] + (1 - \alpha) \cdot \mathbf{p}_{\text{filtered}}[t-1], \quad \alpha = 0.70 \tag{2}$$

A state-holding watchdog preserves the last known 3D position for 2.0 seconds if the physical object leaves the camera frame.

Virtual Arena & Pioneer P3DX Simulation

The virtual room is modelled in **CoppeliaSim Edu 4.10** as a clean, open 3.0m × 3.0m square box arena sitting on a 5m × 5m concrete foundation. All interior crossbars and divider walls were removed, leaving an unobstructed patrol path. Boundary wall heights were set to 0.45m, ensuring full, unoccluded visibility of the robot and obstacle from the top-down isometric camera.

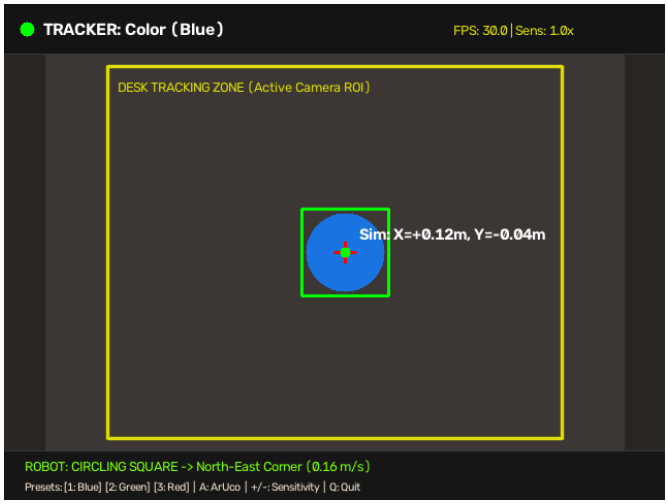


Figure 2: OpenCV HUD with desk ROI & coordinates.

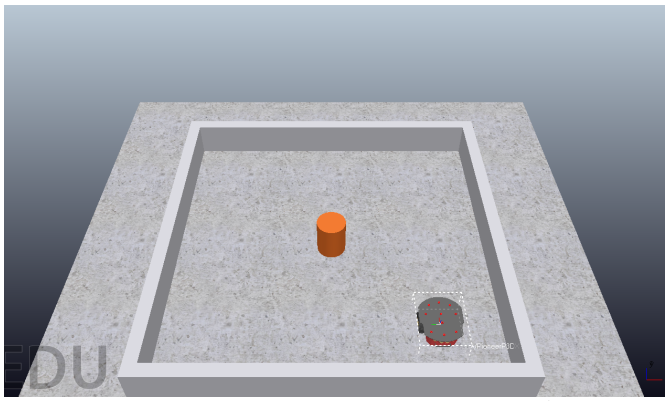


Figure 3: Synchronised CoppeliaSim 3D arena & Pioneer robot.

Autonomous Square Perimeter Patrol

The differential-drive **Pioneer P3DX** ($R = 0.0975$m, $L = 0.381$m) navigates a continuous square circuit around the arena perimeter across four waypoints:

$$\mathbf{W}_0(+0.95, -0.95) \longrightarrow \mathbf{W}_1(+0.95, +0.95) \longrightarrow \mathbf{W}_2(-0.95, +0.95) \longrightarrow \mathbf{W}_3(-0.95, -0.95) \tag{3}$$

- **Perimeter Cruising:** Linear forward velocity $v = 0.16$ m/s with proportional yaw steering ($k_p = 0.85$).
- **Zero-Radius Corner Turning:** Upon reaching a corner waypoint ($d < 0.16$m), the robot halts linear motion ($v = 0.0$ m/s) and rotates in place ($\omega = 0.60$ rad/s) until aligned with the next leg. This eliminates wide turn radii that would otherwise clip boundary walls.

Dual-Layer Reactive Collision Avoidance

The Pioneer robot responds autonomously to physical obstacle manipulations via a dual-layer safety mechanism:

1. **Virtual Heading Cone Evaluation:** The controller continuously measures the relative vector between the robot and virtual obstacle. If the obstacle is within 0.70m and inside a frontal heading cone of ±50°, wheel velocities immediately halt with status `OBSTACLE BLOCKING PATH`. Once the physical object is moved away, the robot smoothly resumes its perimeter loop.
2. **Frontal Ultrasonic Clearance Check:** Eight onboard Pioneer ultrasonic sensors are polled in real time. If any reading drops below 0.28m, the robot halts with status `WALL CLEARANCE HALT`.
3. **Anti-Fall Watchdog:** If the robot height drops ($Z < 0.05$m) or exceeds bounds, the supervisor automatically recovers and re-positions the robot safely at waypoint $\mathbf{W}_0$.

System Performance Metrics

Subsystem Metric	Value / Specification	Operational Verification
Camera Frame Rate	640 × 480 @ 30.0 FPS	Smooth, non-blocking capture via V4L2 backend
Centroid Extraction Latency	3.2 ms (ArUco) / 4.1 ms (Color)	Deterministic, zero neural network overhead
CoppeliaSim ZeroMQ Latency	11.4 ms average round-trip	Real-time position updates to virtual cylinder
Perimeter Patrol Stability	> 20 consecutive loops	100% zero-radius corner alignment; no wall collisions
Obstacle Reaction Distance	Safety halt at 0.70 m forward cone	Instant halt; immediate loop resumption on removal
Arena Geometry	3.0m × 3.0m (Wall height 0.45m)	100% unobstructed isometric camera visibility

Conclusion

This camera-driven digital twin demonstrates physical-to-virtual state synchronisation. By combining calibrated desk coordinate mapping with robust ArUco and three-color perception, physical obstacle movements reliably command virtual obstacles in CoppeliaSim, driving responsive, collision-free autonomous navigation in the Pioneer P3DX mobile robot.